

XFoodRec: An Explainable Mobile Recommender for Personalized Healthy Eating

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Abstract

Food recommender systems often struggle with popularity bias that favors unhealthy recipes, making the promotion of healthier items a significant communication challenge. While explainable recommendations can help justify healthy trade-offs, evaluating their real-world impact is often hindered by the technical complexity of conducting online user studies. We present **XFoodRec**, an extensible mobile research platform designed to bridge the gap between offline algorithmic evaluation and online behavioral studies. The system integrates personalized recipe recommendation with an LLM-based explanation pipeline that translates structured nutritional data into persuasive natural-language explanations. Its built-in A/B testing framework automatically assigns users to experimental groups to measure how explanations influence real-world user behavior and engagement metrics. By providing a modular and deployable infrastructure for controlled experimentation, XFoodRec supports end-users in making healthier meal choices while enabling researchers to investigate how transparency mechanisms influence decision-making in real-world mobile settings. The demonstration video and all code resources are publicly available at: <https://github.com/Amir-Mol/XFoodRec>

CCS Concepts

• Information systems → Language models; • Applied computing → Health care information systems.

Keywords

Recommender system, Food recommendation, Healthy Recommendation, Explainable Recommendation

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1 Introduction

Recommender systems (RS) in the food domain are becoming common tools for everyday meal planning, but they often carry over biases from the data they are trained on. [9]. Previous research [3,

12, 13] has shown that the most popular and highly rated recipes on food social networks tend to be unhealthy. As a result, many existing models prioritize these popular options, which can limit their ability to effectively promote healthy eating [2, 5]. Promoting healthier choices is therefore not only a ranking problem but also a behavioral and communicative challenge [4]. When systems recommend healthier alternatives that are less popular or less aligned with users' preferences, simple prediction is insufficient. If the goal is to influence behavior, users need to understand *why* a particular item is recommended instead of another. Explanations become essential for justifying trade-offs and building trust in system decisions [17].

However, a key open issue remains: how to evaluate the *real effect* of explanations on user decisions [8, 15]. Much existing research [1, 6, 11, 17] relies on offline metrics (e.g., transparency or proxy explanation scores), which cannot directly capture whether explanations increase acceptance, trust, or willingness to try items. In contrast, online evaluation [16], where real users interact with a live system, is considered more reliable for measuring effectiveness and persuasiveness because it observes actual user behavior rather than simulated outcomes. The challenge, however, is that online experiments are costly and technically demanding: they require a fully implemented mobile or web platform, user interaction logging, controlled experimental design, and deployment infrastructure [11].

To address this gap, we present *XFoodRec*, an explainable mobile food recommender designed not only as a healthy-eating assistant but also as a research platform for online evaluation of explainable recommendation strategies. The system integrates structured nutritional reasoning, personalized user profiling, and LLM-based explanation generation within a deployable Flutter–FastAPI mobile architecture. Each user is automatically assigned to either a **Transparency** condition, where recommendations are accompanied by natural-language explanations and health-related reasoning, or a **Control** condition, where identical recommendations are shown without explanations. This built-in A/B testing framework allows controlled, real-world experimentation on how transparency influences perceived justifiability and acceptance of healthier items.

Beyond its role as a healthy-eating assistant, *XFoodRec* is designed as a reusable experimental infrastructure for explainable RS. Its modular architecture allows researchers to replace recommendation/explanation modules while reusing the same mobile interface, feedback pipeline, and built-in A/B testing framework. By embedding controlled online experimentation directly into a deployable mobile app, XFoodRec lowers the technical barrier to studying how transparency affects user decisions, helping bridge the gap between algorithmic research and real-world behavioral evaluation. Main contributions of this study are:



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- **Deployable mobile experimentation platform.** A functional app supporting controlled A/B comparison of recommendations with and without explanations.
- **Modular LLM-based explainability.** Personalized explanation generation and easy module replacement.
- **Built-in behavioral instrumentation.** Integrated logging of interaction signals for online evaluation.
- **Reusable research testbed.** An extensible infrastructure designed for real-world studies on transparency in RS.

2 System Overview

XFoodRec is a mobile platform that integrates structured nutritional data, personalized user profiling, and LLM-based reasoning to deliver explainable healthy food recommendations. The architecture (Figure 1) consists of three main modules: (i) user onboarding and profile management, (ii) a two-stage hybrid recommendation engine, and (iii) an explainability layer and user feedback collection. The system was built to be both computationally efficient for mobile environments and flexible enough to support A/B testing and experimental research on explainable recommendations.

2.1 Data and User Profiling

The system relies on a structured recipe dataset of HUMMUS dataset [2]) that includes food names, images, descriptions, ingredients, and detailed nutritional values (calories, macronutrients, vitamins, and minerals). Recipes are vectorized for efficient similarity computation. During onboarding, users complete a multi-step profiling process, including: **Basic Profile** (e.g., age, gender, height, weight, and activity level), **Dietary Needs** (e.g., structured data representing allergies, intolerances, or health conditions such as diabetes and hypertension) and **Taste Profile** (e.g., preferred cuisines, liked and disliked ingredients, and flavor preferences).

The profile is continuously refined through a feedback loop, where users rate recommendations and express *likes*, *dislikes*, or *willingness-to-try responses*. These behavioral data are recorded for the retraining and statistical evaluation of the experiment.

2.2 Recommendation Engine

At the core of XFoodRec is a two-stage hybrid model that combines deterministic filtering with generative reasoning:

Stage 1 - Filtering Engine: This module prunes the master recipe dataset into a personalized consideration set (100 candidates). **Hard Constraints** (“Never List”) eliminate recipes that violate user restrictions (e.g., allergies or vegan requirements). **Soft Constraints** (“Influence Zone”) generate an initial ranking using cosine similarity between user and recipe embeddings. We employ the `thenlper/gte-small` dense embedding model (a lightweight BERT-based encoder) to project both user profiles and recipe meta-data into a shared 384-dimensional vector space. Recipe representations are constructed by concatenating title, ingredients, tags, and nutritional attributes, while user representations combine health conditions, dietary preferences, and activity levels.

Rather than applying a fixed similarity threshold, we adopt a dynamic Top-K strategy, retrieving the top $K = 100$ candidates with the highest adjusted similarity scores. The base similarity is further refined using multiplicative heuristics: (i) a 20% bonus for preferred

ingredients and a 70% penalty for disliked ones; (ii) health-aware penalties (e.g., 50% penalty for recipes exceeding sugar limits for diabetic users, or 40% for high sodium in hypertensive users); and (iii) a 30% penalty for recipes falling outside a $\pm 50\%$ range of the user’s estimated per-meal calorie target.

Stage 2 – Generation Engine: The filtered set is re-ranked by an LLM-powered module that also produces natural-language explanations. Using OpenAI GPT-4o, the model acts as a virtual nutritionist, reasoning over candidate recipes and outputting JSON-formatted responses with recipe IDs, rankings, and concise explanations covering taste, nutrition, and drawbacks.

This two-stage design achieves both speed (via efficient vector filtering) and depth, suitable for interactive mobile scenarios.

2.3 Explainability Layer

The explainability component transforms model outputs into transparent user-facing narratives. Each explanation is guided by structured prompt templates that ensure coverage of three aspects: 1- **Taste Alignment:** how the recommendation matches user preferences, 2- **Health Context:** nutritional reasoning and relevance to dietary goals and 3- **Drawbacks or Caveats:** explicit mention of limitations, such as high sodium or sugar content. Also, two versions of the interface are deployed for A/B testing: 1- the *Transparency UI*, which displays the full explanation and reasoning trace; and the *Control UI*, which presents recommendations without explanations. This mechanism supports controlled experiments and the design of user studies.

2.4 Implementation

Mobile Application. The Android app can be easily sideloaded on standard Android devices for live demonstration and participant testing. Its UI/UX design follows Google Material Design 3 principles, emphasizing clarity, accessibility, and minimal cognitive load. Core screens, onboarding, recommendation feed, and feedback dialog, were implemented using the Flutter framework.

Frontend Development. The user interface and experience are built using Material Design 3 components and typography, ensuring a clean and consistent visual identity. The user flow includes multi-step onboarding, a dynamic profile editor, and interactive feedback widgets to enhance engagement. A/B testing logic is integrated directly into the recommendation loop, automatically assigning users to either Control or Transparency groups for experimental evaluation. Additionally, the feedback system captures key user interactions such as likes, dislikes, and healthiness ratings, enabling continuous refinement of personalization models.

Backend Development. The backend is implemented using FastAPI (Python) for scalable API endpoints, supported by the Prisma Client Python ORM and a PostgreSQL database for efficient data management.

includes the complete system implementation, covering back-end services, frontend mobile application, recommendation engine, LLM-based explanation pipeline, evaluation scripts, and prompt templates used for LLM reasoning and assessment. We also provide representative Stage 2 prompts for LLM-based re-ranking and explanation generation in the repository. Data collection is structured within a dedicated *TrainingRecord* table that stores user group

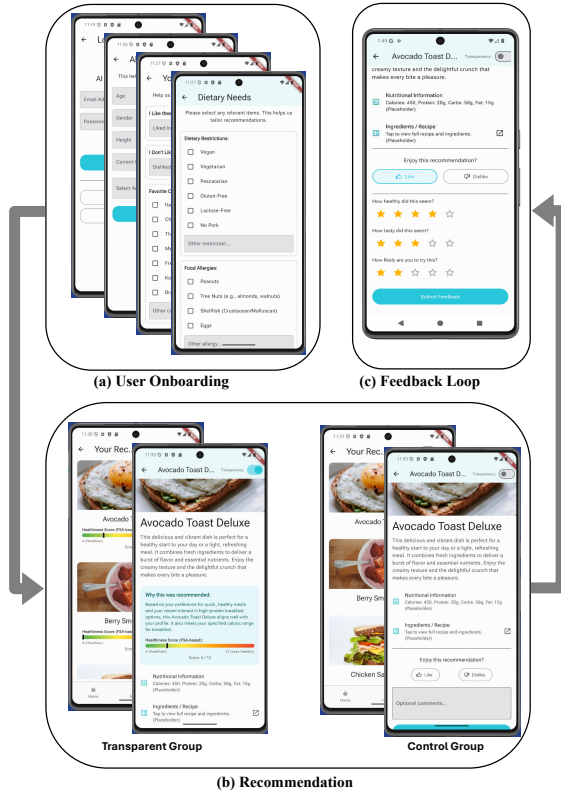


Figure 1: Explainable recommendation user journey.

assignments, session identifiers, and detailed feedback metrics, enabling subsequent statistical and behavioral analysis.

3 Explainable Recommendation Interface

The XFoodRec mobile interface enables users to receive, understand, and evaluate personalized meal suggestions through a transparent, easy-to-use workflow. It emphasizes clarity and accessibility for both daily use and research reproducibility. The end-to-end interaction (Figure. 2) includes user onboarding, personalized recommendation delivery, and feedback submission, turning explainability into a continuous, user-centered experience.

3.1 Scenario Overview

Users enter demographic and dietary preferences that form their personal profile. Once completed, the system automatically assigns each user to one of two experimental conditions:

Transparency Scenario. In the Transparency View (Figure 2(b)), the recipe profile, comprising the image, ingredients, and nutritional facts, is displayed alongside a section titled “*Why this was recommended*”, generated by the LLM module. Each explanation highlights preference alignment, nutritional reasoning, and relevant constraints, for example: “*Based on your preferences for quick, healthy meals and your recent interest in high-protein breakfast options, this Avocado Toast Deluxe aligns well with your profile.*” A visual **Health Score** complements the text.

Control Scenario. The Control View (Figure. 2(b)) presents the same layout of the recipe profile without explanations or health score, isolating the effect of transparency.

Both interfaces include identical feedback options (*Like*, or *Dislike*) and optional comments.

This A/B structure allows measuring the effect of explanations.

Feedback collection. Feedback is logged for post-hoc statistical analysis to assess whether the presence of explanations increases the likelihood of selecting healthier food options.

4 Evaluation

Evaluating the *real behavioral impact* of healthy food recommendations and explanatory transparency requires longitudinal user studies, where sustained dietary choices are observed over time to detect behavior change. This mobile app is primarily designed as a **research platform** for future explainable recommendations: researchers can replace the recommendation engine and explanation generation modules while reusing the same mobile interface, logging infrastructure, and A/B testing framework. This design facilitates rapid prototyping, comparative evaluation, and user-centered A/B analysis across different recommendation and explanation strategies. Due to time constraints, we focus here on a complementary step, **technical validation** of the recommendation engine and explainability layer, rather than long-term behavioral analysis.

4.1 Evaluation Design

We constructed $N = 10$ synthetic user profiles to reflect diverse and realistic dietary situations, including combinations of: (i) allergies (e.g., nuts, lactose), (ii) dietary styles (e.g., vegetarian), (iii) health conditions, and (iv) diverse taste preferences and cuisines. Each profile contains the same structured fields collected in XFoodRec onboarding (demographics, dietary needs, and taste preferences).

We evaluate the system using an instance-level A/B setup: **Condition A (Control)**: recommended items and their ingredients and nutritional facts, *without* explanations. And **Condition B (Transparency)**: recommended items and their ingredients and nutritional facts, *with* explanations.

For each user profile, we generate three independent recommendation instances per condition ($3 \times A$ and $3 \times B$), obtained through stochasticity in the LLM re-ranking stage. This results in multiple recommendation–explanation pairs per profile, allowing us to assess robustness across generations. We address three RQs. (**RQ1**) Does adding explanations improve perceived Transparency and Persuasiveness compared to the Control condition? (**RQ2**) Do A/B differences across all measures vary between Nutritionist and RS Experts roles? (**RQ3**) To what extent do LLM-based evaluators align with human experts under identical role definitions?

4.2 Evaluators

After generating recommendations and explanations using our recommendation and explanation engines, we evaluate them with both human experts and LLM-based evaluators operating under identical expert roles. Human evaluations serve as the primary source of evidence, while LLM-based evaluators, implemented using a language model different from the one used in the recommendation engine, provide scalable, role-aligned assessments that help

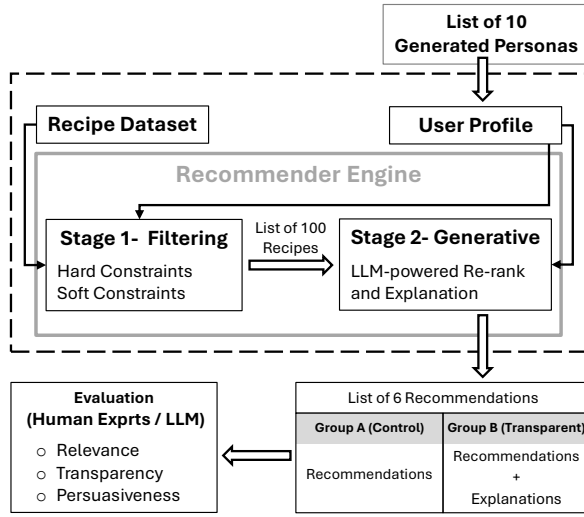


Figure 2: Explainable RS evaluation design.

identify inconsistencies and support large-scale analysis. Recent research has shown that LLM can function as reliable “LLM-as-a-judge” evaluators when guided by clearly defined rubrics, often demonstrating meaningful correlation with human judgments for subjective evaluation tasks such as explanation quality and reasoning assessment [7]. When used in parallel with human experts and under the same role definitions, LLM-based evaluation can improve consistency, scalability, and reproducibility, while humans remain the gold standard for final interpretation. To ensure role symmetry and fair comparison, both human and LLM evaluators operate under two expert perspectives:

Nutritionist Role: The Human Nutritionist (H1) and the LLM-based evaluator operating in the Nutritionist role (L1, Gemini-2.5-Flash) assess nutritional correctness, health suitability, dietary risks, and the consistency of nutritional reasoning.

RS Expert Role: The Human RS expert (H2) and the LLM-based evaluator operating in the RS role (L2, Gemini-2.5-Flash) assess the alignment of recommendations with user preferences and tastes.

Evaluators are provided with the user profile, the recommended items and their nutrition facts, and (only for Condition B) the explanation text. To reduce bias, the order of A vs. B presentation is randomized and the condition label is hidden.

4.3 Evaluation Criteria

All subjective ratings are provided by evaluators operating under two roles: RS Expert and Nutritionist. Each metric is rated on a 1–5 Likert scale. For each user–recommendation pair (and the associated explanation in the Transparency group), the evaluator reports:

-Relevance aims to evaluate “*whether the recipe is suitable for the user’s dietary profile and preferences, considering dietary restrictions, allergies, and health conditions*”.

-Transparency: aims to evaluate “*whether the explanations can reveal the internal working principles of the recommender model*” [10].

-Persuasiveness: aims to evaluate “*whether the explanations can increase the interaction probability of the users on the items*” [14].

4.4 Results

RQ1 (Effect of Explanations). Mann–Whitney U tests reveal a strong and statistically significant improvement in both perceived Transparency and Persuasiveness in the Transparency condition compared to Control. Overall, median Transparency increased from 1.0 in Control to 4.0 in Treatment ($p < 10^{-19}$), while Persuasiveness increased from 1.0 to 3.5 ($p < 10^{-9}$). The effect is consistent across all evaluators and remains significant under persona-level aggregation. These results indicate that adding natural-language explanations substantially enhances perceived transparency beyond presenting ingredients and nutritional facts alone.

RQ2 (Role-dependent effects). Across both evaluator roles, the Transparency condition significantly increased perceived Transparency and Persuasiveness, while Relevance remained unaffected. At the persona-aggregated level, both recommendation Expert and Nutritionist showed significant gains in Transparency ($p < 0.001$) and Persuasiveness ($p < 0.05$), confirming robustness across roles. However, difference-of-differences analysis reveals that the increase in Persuasiveness is significantly larger for Recommendation Experts than for Nutritionist ($p = 0.022$), whereas improvements in Transparency do not differ significantly between roles. These findings suggest that explanations consistently enhance perceived clarity across expert perspectives, but their persuasive impact is more pronounced among AI-focused evaluators.

RQ3 (Human vs. LLM Alignment). Spearman correlation analysis reveals role-dependent agreement between human and LLM-based evaluators. Alignment is strongest for Transparency, with significant positive correlations in both the Nutritionist role ($\rho = 0.41$, $p < 0.01$) and Recommendation Expert role ($\rho = 0.63$, $p < 10^{-7}$), and a pooled correlation of $\rho = 0.53$. Persuasiveness shows moderate alignment, particularly in the Recommendation Expert role ($\rho = 0.41$, $p < 0.01$). In contrast, Relevance exhibits no significant correlation in either role ($\rho \approx 0$), indicating weak agreement on intrinsic recommendation suitability. These findings suggest that LLM-based evaluators show moderate agreement with human judgments for explanation-related dimensions (transparency and persuasiveness), but are less consistent in assessing core recommendation relevance. A summary of all examined RQs is presented in Table 1.

4.5 Limitations

Our evaluation in this study focuses on expert-based and LLM-based technical validation rather than real user behavior analysis. Ratings were obtained from a limited number of synthetic personas and four evaluators, which may not capture the variability of real-world populations. While LLM-based judges enable scalable analysis, their partial misalignment with human experts—particularly for relevance judgments—indicates that automated evaluation cannot fully replace human assessment. This limitation becomes more pronounced in clinically complex scenarios (e.g., Persona 2: a Type 1 diabetic nurse with a severe tree-nut allergy), where stricter medical reasoning by human experts leads to lower relevance scores, highlighting the need for domain-expert oversight in high-risk dietary settings. Furthermore, the current prototype relies on GPT-4o for re-ranking and explanation generation, introducing potential

Table 1: Summary of RQ1–RQ3 findings.

RQ	Metric	Key Result
RQ1	Transparency	Median: 1.0 $\rightarrow$ 4.0 ($p < 10^{-19}$)
	Persuasiveness	Median: 1.0 $\rightarrow$ 3.5 ($p < 10^{-9}$)
RQ2	Transparency	Significant increase for both roles ($p < 0.001$)
	Persuasiveness	Larger gain for RS Experts ($p = 0.022$)
	Relevance	No significant A/B difference
RQ3	Transparency	Moderate–strong human–LLM agreement ($\rho = 0.53$ pooled)
	Persuasiveness	Moderate agreement (RS role $\rho = 0.41$)
	Relevance	No significant correlation ($\rho \approx 0$)

challenges related to reproducibility, cost, and dependency on proprietary models; however, the modular design allows integration of local or open-source alternatives in future work. Finally, the study evaluates perceived transparency and persuasiveness but does not assess long-term behavioral change or real-world decision-making. Ongoing work includes a small-scale pilot study with real users to address these limitations.

5 Conclusion

XFoodRec demonstrates how explainable AI can enhance transparency and engagement in personalized healthy eating recommendations. By integrating nutritional reasoning and LLM explanations, it supports both practical use and controlled behavioral research. The built-in A/B framework enables rigorous comparison between transparency and control conditions, providing a deployable research platform for studying personalized and explainable food recommendations in real-world settings. Ongoing work includes a small-scale pilot with real users to evaluate usability, recommendation quality, and explanation effectiveness; thus, current results should be interpreted as preliminary technical and expert-based validation rather than evidence of long-term behavioral change.

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